

Wired

Wired headphones carry audio signals in well insulated copper cables and connect through the analog (or digital) ports to a device. These analog ports can be the 3.5mm/6.25mm audio jack (R.I.P. Headphone jack) or the RCA ports which use separate ports for the Left and the Right channel. The 3.5mm jack has evolved over the years to support stereo sound and even integrates electronics for remote control and microphone inputs. Headphone jacks are however, analog and are being slowing and quite reluctantly phased out in favour of digital ports. Digital ports are in the likes of the microUSB, USB-C or lightning ports on the devices and some even transmit thought dedicated Optical cables. For a deep dive into ports flip to Chapter 8.

Wireless

Wireless headphones on the other hand do not rely on the device to do the processing. The bits and bytes are directly transmitted over bluetooth, RF or WiFi to the device and the DAC on the headphones converts the signals to analog for hearing. Since wireless headphones integrate so much tech and electronics into the ear cups they often compromise on the sound quality. There are also some signal losses when transmitting data wirelessly and the DAC has to compensate for that as well. The headphones also need to have integrated batteries and power circuits to keep the headphones tether free and the drivers are often smaller than those on wired headphones to



Truly wireless headphones are on the rise